



## OPEN Impact of bed roughness configurations on flow dynamics and hydraulic resistance in open-channel flows

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Bed roughness significantly influences open-channel flow dynamics, affecting velocity profiles, energy distribution, and hydraulic resistance in natural and engineered rivers. This study investigates the effects of four bed roughness configurations, smooth forward (S-FW), offset roughness (R-OFF), and forward-oriented roughness at 60° (R-FW60) and 90° (R-FW90), on flow characteristics in an experimental flume. Velocity measurements, energy calculations, spectral analysis, and statistical tests (ANOVA) were employed to quantify the impact of roughness on mean flow velocity, Chezy coefficient ( $C$ ), energy dissipation, and turbulence. Results reveal that S-FW yields the highest mean velocity (1.1587 m/s) and  $C$  (51.62), indicating minimal resistance, while R-FW90 produces the lowest mean velocity (0.4422 m/s) and  $C$  (18.51), reflecting significant drag and turbulence. Energy profiles show pronounced dissipation near rougher beds, with R-FW90 exhibiting the lowest near-bed energy due to vortex shedding. Spectral analysis highlights amplified low-frequency velocity fluctuations in rougher configurations, particularly R-FW90. Linear mixed-effects modeling and post-hoc tests confirms significant velocity differences across configurations ( $p < 0.05$ ). These findings offer insights for optimizing channel design, managing erosion, and enhancing flow efficiency in hydraulic engineering applications.

**Keywords** Bed roughness, Open-channel flow, Velocity profiles, Chezy coefficient, Energy dissipation, Turbulence, Flume experiment

### List of symbols

|                     |                                                           |
|---------------------|-----------------------------------------------------------|
| $B$                 | Channel width (m)                                         |
| $C$                 | Chezy coefficient ( $\text{m}^{1/2}/\text{s}$ )           |
| $C_{\text{rough}}$  | Chezy roughness coefficient ( $\text{m}^{1/2}/\text{s}$ ) |
| $C_{\text{smooth}}$ | Chezy smooth coefficient ( $\text{m}^{1/2}/\text{s}$ )    |
| $E_k$               | Kinetic part of specific energy (m)                       |
| $F$                 | Gravitational acceleration (–)                            |
| $g$                 | Gravitational acceleration ( $\text{m}/\text{s}^2$ )      |
| $h_L$               | Energy loss between consecutive positions (m)             |
| $K$                 | Height of roughness elements (m)                          |
| $R$                 | Hydraulic radius (m)                                      |
| $Re$                | Reynolds number (–)                                       |
| $S$                 | Distance between roughness elements (m)                   |
| $S_o$               | Channel bed slope (m)                                     |
| $t$                 | Thickness of roughness elements (m)                       |
| $V_{\text{ave}}$    | Measured velocity at vertical position (m/s)              |
| $We$                | Weber number (–)                                          |
| $Y$                 | Flow depth (m)                                            |
| $z$                 | Bed elevation (m)                                         |

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|            |                                                                   |
|------------|-------------------------------------------------------------------|
| $\Delta z$ | Spacing between measurement points (m)                            |
| $\mu$      | Dynamic viscosity (Pa s)                                          |
| $\rho$     | Water density (Kg/m <sup>3</sup> )                                |
| $\sigma$   | Surface tension (N/m)                                             |
| $\theta$   | Angle of roughness strips relative to the flow direction (degree) |

Rivers have long been the lifelines of the Earth, not only transporting water, nutrients, and sediments across natural landscapes but also serving as arteries of human civilization, playing an indispensable role in societal development. Despite the significant threats facing natural river waters, as noted by Abbott<sup>1</sup>, humans have consistently regarded these waterways and river valleys as vital sources for drinking water, agriculture, and industry, reliable transportation routes, and sites for energy production through hydropower<sup>2</sup>. More than one billion people live near rivers<sup>3</sup>, and the riverine areas, due to their natural attractions and recreational and tourism opportunities, have become ideal locations for urban settlement<sup>4</sup>.

Rivers are also dynamic and fragile ecosystems, whose delicate balance is influenced by hydro-morphological factors and human interventions<sup>5,6</sup>. This dynamism makes the study and modeling of river flows one of the most complex challenges in hydraulic engineering, where hydraulic resistance<sup>7</sup> emerges as a key concept at the heart of these challenges<sup>8</sup>. Yet beneath this vital role lies a persistent engineering challenge. In steep mountainous rivers with gradients exceeding 1–3%, intense sediment transport and rapid reservoir infilling threaten infrastructure longevity and water security. In such environments hydraulic resistance becomes a controlling factor that governs flow velocity, energy dissipation, and sediment dynamics, making it a central element in river engineering.

Experimental investigations into the influence of artificial bed roughness on hydraulic resistance date back to the seminal work of Adachi<sup>4,9</sup>, who conducted controlled flume experiments using artificial strip roughness elements aligned perpendicular to the flow. His results demonstrated that hydraulic resistance transitions from isolated roughness ( $s/k > 40$ ) to wake-interference ( $8 < s/k < 40$ ) and finally to skimming flow ( $s/k < 8$ ), with peak resistance occurring at  $s/k = 8$ . These findings established a foundational framework for understanding roughness spacing effects and remain a benchmark in open-channel hydraulics. Subsequent studies extended this concept to natural and quasi-natural beds. Türker and Valyrakis<sup>10</sup> investigated gravel-bed channels with randomly distributed clasts and showed that changes in protrusion orientation can strongly modify near-bed turbulence and the thresholds for particle entrainment. Nepf<sup>11</sup> evaluated drag partitioning in vegetated channels and demonstrated that the alignment of stems plays an important role in controlling form drag and the development of secondary currents. Stewart et al.<sup>6</sup> examined self-affine bed topographies and reported that fractal roughness creates spatial variability in hydraulic resistance. More recently, Coscarella et al.<sup>12</sup> and Zhang et al.<sup>13</sup> investigated 3D bump-like protrusions, confirming that angled or offset roughness generates asymmetric vortex streets and enhances energy dissipation compared to transverse bars.

To better understand the underlying processes, it is essential to recognize that hydraulic resistance, the force that impedes the movement of flowing water, plays a central role in the analysis and design of riverine systems. This concept is widely applied in the mathematical modeling of free-surface flows, particularly within the framework of one-dimensional (1D) and two-dimensional (2D) shallow water models based on the nonlinear Saint-Venant equations<sup>14</sup>. These models, implemented in advanced computational tools such as HEC-RAS, allow for the analysis of water surface profiles, sediment transport, and water quality under both steady and unsteady conditions<sup>15</sup>. However, complexities arising from bed irregularities, such as ridges, reefs, vegetation, suspended sediments, ice, and channel geometry variations, make hydraulic resistance a highly variable and challenging phenomenon<sup>11,16</sup>. These hydro-morphological factors, which can change randomly, seasonally, or locally<sup>6</sup>, can lead to significant variations in water levels and flow discharge, for example, in channel constrictions caused by ice jams, debris accumulation, or alluvial sediment deposits<sup>17</sup>. Such variations can complicate accurate flow prediction and flood risk management, as errors in estimating hydraulic resistance may result in catastrophic consequences<sup>18,19</sup>. In this context, shallow water models with advanced hydrodynamic solutions, derived from the Navier–Stokes equations and Reynolds-averaged Navier–Stokes (RANS) formulations, serve as powerful tools for simplifying these complex problems and providing precise boundary conditions<sup>20</sup>.

Along this winding path, rivers serve not only as vital resources for life and development but also as natural laboratories for studying open-channel flow dynamics. Bed shear stress, as a key parameter in river engineering, plays a decisive role in the interaction between water and the riverbed, shaping flow behavior and boundary conditions<sup>21</sup>. The presence of bed irregularities, such as three-dimensional protrusions or roughness elements oriented perpendicular to the flow, intensifies flow turbulence and increases shear forces<sup>10,12</sup>. These disturbances not only accelerate bed and bank erosion but also threaten ecological balance and adjacent infrastructure<sup>22</sup>. Researchers, by studying simplified models such as bump-like protrusions, have been able to analyze the effects of these irregularities on hydraulic parameters including head loss, turbulent flow structures, and velocity distribution<sup>13,23–25</sup>. These findings, rooted in decades of hydraulic engineering research<sup>26,27</sup>, have provided valuable tools for designing water conveyance systems, sediment control, and erosion mitigation.

At the core of these analyses, the Chezy coefficient ( $C$ ) emerges as a key empirical parameter that estimates the mean flow velocity and plays an indispensable role in hydraulic design<sup>28,29</sup>. Rooted in Antoine de Chezy's pioneering work in 1775, this coefficient integrates the effects of bed roughness, channel cross-sectional shape, and flow conditions, providing a powerful tool for predicting flow and energy losses. Bazin<sup>30</sup> took a significant step toward operationalizing it by demonstrating its dependence on the hydraulic radius and bed texture, while Carlier<sup>31</sup> emphasized the role of surface roughness. Powell<sup>32</sup> advanced the concept by linking the Chezy coefficient to roughness scale and flow intensity, and Swamee and Rathie<sup>33</sup> contributed an explicit formula incorporating viscosity and slope. Nevertheless, modeling challenges in channels with complex geometries remain, as highlighted by Loukam and colleagues<sup>34,35</sup>, who stress the need for multipurpose equations for diverse configurations<sup>36–41</sup>.

This study builds upon controlled experimental tradition established by Adachi<sup>4,9</sup> by exploring oriented bar roughness, contributing to a better understanding of roughness effects on flow resistance and turbulence.

and extending it into the previously unexplored domain of oriented bar roughness, thereby providing the missing link between laboratory-scale turbulence control and field-scale sediment management. Despite extensive research on the effects of bed roughness on open-channel flow dynamics, there is still a limited understanding of how specific roughness configurations influence velocity profiles, energy dissipation, and turbulence characteristics in a comprehensive manner. This study seeks to address these gaps by systematically analyzing the interplay between bed roughness, flow resistance, and energy distribution. The main aims of this research are: (i) to quantify the impact of varying bed roughness configurations (smooth, offset, and angled protrusions) on mean flow velocity and the Chezy coefficient; (ii) to evaluate energy distribution and dissipation patterns across different roughness setups; (iii) to investigate velocity fluctuations through spectral analysis to characterize roughness-induced turbulence; (iv) to assess the statistical significance of roughness effects on flow dynamics using ANOVA and advanced visualization techniques; and (v) to provide practical insights for optimizing channel design and managing erosion in hydraulic engineering applications.

### Non-dimensional parameter analysis

The velocity field in open-channel flows is controlled by a combination of hydraulic and geometric variables, such as discharge, fluid viscosity coefficient ( $\mu$ ), acceleration due to gravity ( $g$ ), interfacial tension ( $\sigma$ ), fluid mass density ( $\rho$ ), spacing between neighboring roughness elements ( $S$ ), element height ( $K$ ), longitudinal bed gradient ( $S_0$ ), bulk velocity ( $V$ ), channel span ( $B$ ), and strip thickness ( $t$ ). The overall dependence can be expressed in the general form:

$$F(\mu, g, \sigma, \rho, S, K, S_0, Y, V, B, t, \text{Roughness Configuration}) = 0 \quad (1)$$

By applying dimensional analysis, these parameters are transformed into dimensionless groups as:

$$\frac{V}{\sqrt{RS_0}} = (\text{Re}, \text{We}, \text{Fr}, S_0, S/K, t/K, B/K, B/S, \text{Roughness Configuration}) \quad (2)$$

Where the Reynolds number is defined as  $\text{Re} = \frac{\rho V Y}{\mu}$ <sup>41–47</sup>, and the Weber number as  $\text{We} = \frac{\rho V^2 Y}{\sigma}$ <sup>48–50</sup>. Here,  $S_0$  denotes the channel gradient,  $S/K$  represents the ratio of spacing to element height, and  $Y$  is the flow depth. The term  $\frac{V}{\sqrt{RS_0}}$  is equivalent to the Chezy coefficient, where  $R$  is the hydraulic radius defined as  $R = A/P$ , with  $A$  being the cross-sectional flow area and  $P$  the wetted perimeter. Although the Chezy coefficient is not dimensionless, it is retained in the analysis because of its fundamental importance in hydraulic engineering and its direct relevance to characterizing resistance in open-channel flows. In uniform flow conditions, such as in the present experiment, the friction slope equals the bed slope ( $S_f = S_0 = 0.02$ ).

In this study, the physical characteristics and arrangement of the roughness strips were kept constant. The selected values were  $S_0 = 2\%$ ,  $t/K = 1$ ,  $B/K = 14.28$ ,  $B/S = 2.5$ , and  $S/K = 5.71$ . Because the flow was fully turbulent and surface tension effects were negligible, the Weber number was not considered. Similarly, the Reynolds number remained unchanged across tests. Consequently, Eq. (2) simplifies to:

$$C = (\text{Fr}, \text{Roughness configuration}) \quad (3)$$

In turbulent flows over rough beds, however, the friction slope can be expressed using the Darcy–Weisbach formulation:

$$S_f = \frac{fV^2}{8gR} \quad (4)$$

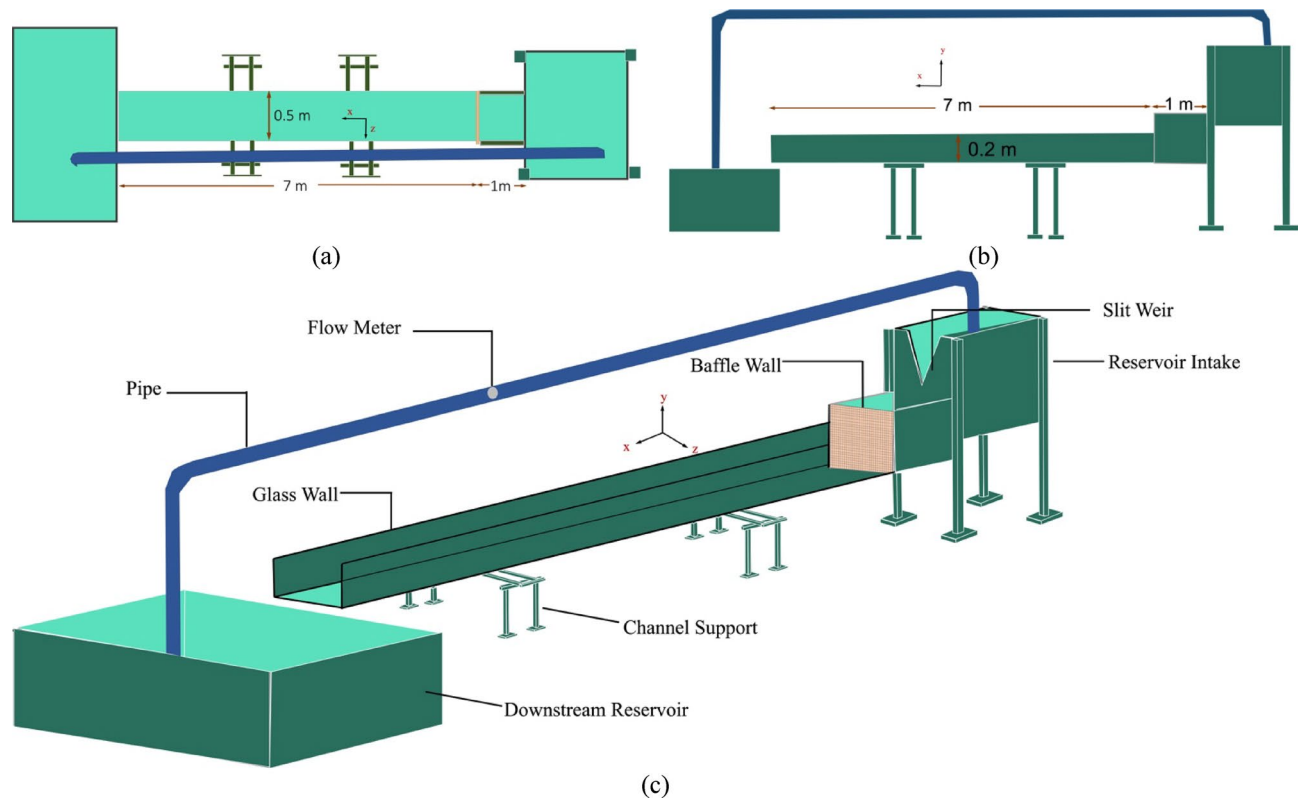
where  $f$  is the Darcy–Weisbach friction factor and  $g$  is gravitational acceleration. Substituting this into the Chezy relation yields:

$$S_f = \sqrt{\frac{8g}{f}} \quad (5)$$

The friction factor  $f$  depends on the relative roughness, roughness density (spacing of roughness elements), and the Froude number. In this study, the roughness density and  $\text{Fr}$  were kept constant for different models, while only the roughness arrangement was varied.

### Experimental setup

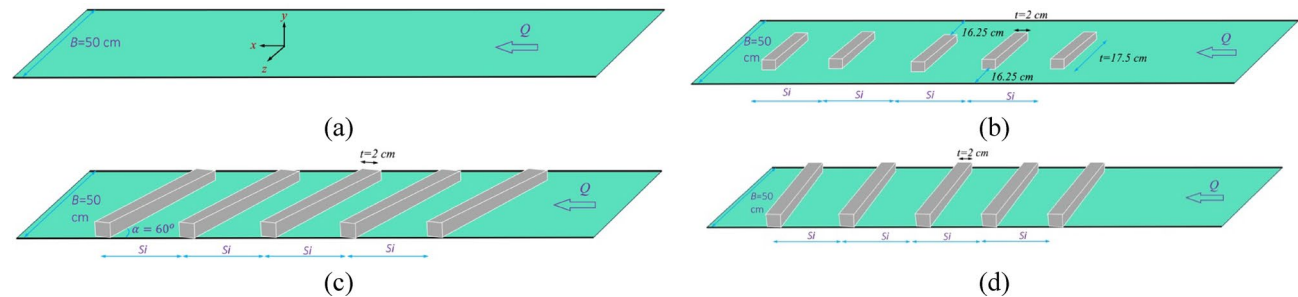
At Kyoto University's Ujigawa Hydraulic Laboratory in Japan, experiments were conducted in a closed-circuit rectangular flume. The channel, constructed from Plexiglas, has dimensions of 0.5 m wide and 7 m long, with a longitudinal slope fixed at 2% (Fig. 1). A pump, regulated by a Mitsubishi FREQROL-A700 variable-frequency drive, circulated water from a constant-head tank through the system, and a calibrated gate valve enabled precise control of the flow rate. The laboratory maintained a consistent air temperature of 21 °C during all experiments. The discharge was set at  $Q = 15 \text{ L/s}$  for all tests.



**Fig. 1.** Schematic diagrams of the experimental flume: (a) plan view, (b) side elevation, and (c) isometric representation.

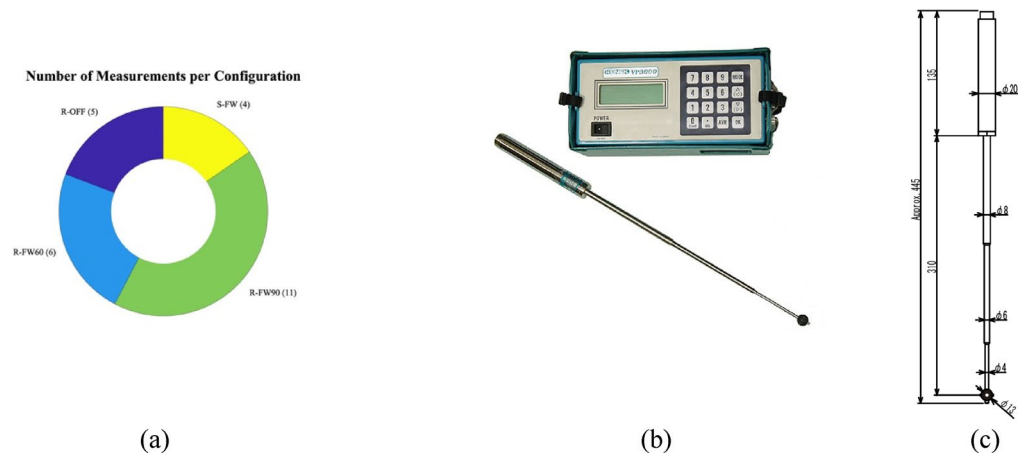
| Case | Bed type | Space (S) | Rough position | Lable  | Description                                                                 |
|------|----------|-----------|----------------|--------|-----------------------------------------------------------------------------|
| 1    | Smooth   | –         | –              | S-FW   | Smooth bed                                                                  |
| 2    | Rough    | 20 cm     | 17.5 cm        | R-OFF  | Rough bed, bar offset 16.25 cm from edge                                    |
| 3    | Rough    | 20 cm     | Full width     | R-FW60 | Rough bed, bar spanning full channel width, at 60° relative to channel flow |
| 4    | Rough    | 20 cm     | Full width     | R-FW90 | Rough bed, bar spanning full channel width, at 90° relative to channel flow |

**Table 1.** Experimental configurations of bed roughness, including smooth and rough channel beds with varying bar positions and orientations.

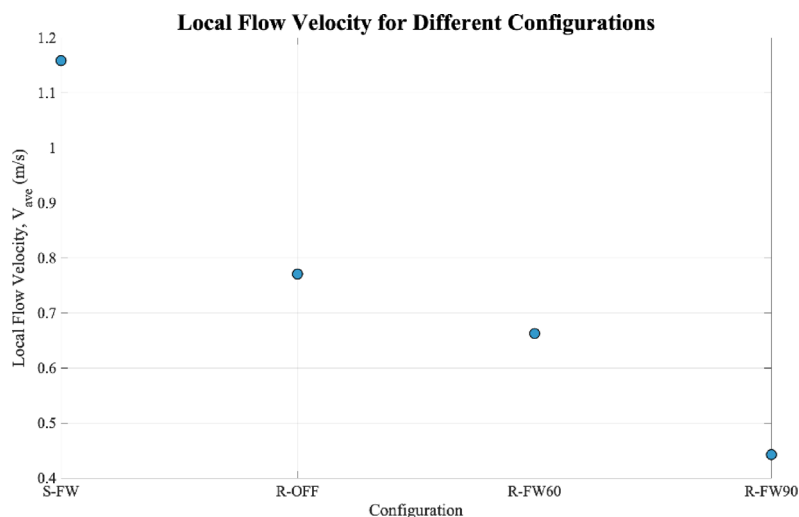


**Fig. 2.** Spatial distribution of roughness elements along the channel bed.

The study involved four distinct channel-bed configurations, S-FW, R-FW60, R-FW90, and R-OFF, as detailed in Table 1. Figure 2 illustrates the arrangement of roughness elements, while Fig. 3a shows the number of measurements taken for each setup. Velocity data collection followed the same methodology as used for the smooth-bed configuration. Velocity measurements were taken at a fixed streamwise location positioned midway between two consecutive roughness strips and at the channel’s transverse midpoint ( $B/2$ ), where  $B$  denotes



**Fig. 3.** (a) Number of experimental runs conducted in this study (b) Kenek flow detector (c) Velocity meter used for point-velocity measurements.



**Fig. 4.** local flow velocity at different points and conditions.

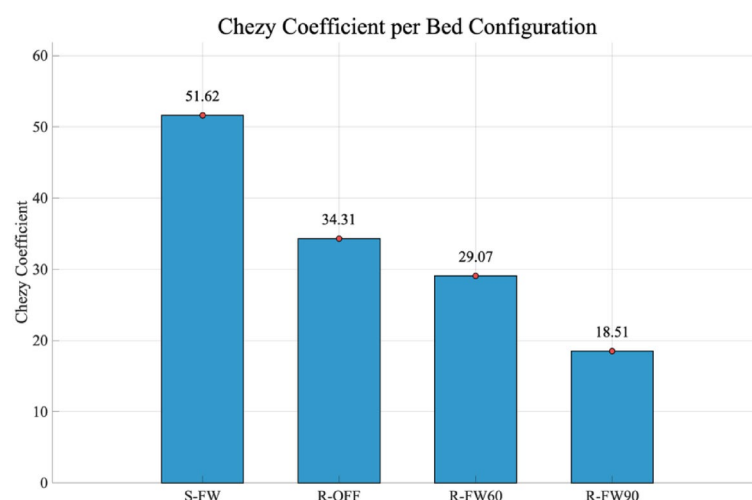
the channel width. All measurements were conducted at this single fixed cross-section to ensure consistency across configurations; longitudinal measurements along the channel were not performed, as the focus was on vertical effects and roughness configuration under steady, uniform flow. Importantly, the measurement points were located above the bed surface, in order to avoid geometric interference with the sensors. This position was kept identical for all roughness configurations and was also replicated for the smooth-bed case at the same longitudinal station. At each measurement point, a Kenek VP3000 electromagnetic flowmeter (Fig. 3b, and c) captured 60 samples of the three velocity components at a 100 Hz sampling rate. To ensure data consistency and repeatability, measurements were duplicated for each point. The minimum measurement height of  $z = 10$  mm is dictated by the physical design of the Kenek VP3000 electromagnetic current meter. The probe has a sensing head (tip) approximately 10 mm in diameter that must be fully submerged in water for the electromagnetic field to function correctly (Fig. 3c). This is a fundamental limitation of electromagnetic current meters in shallow flows and is documented in the instrument manual.

Flow depth was measured at the test section's center using a digital point gauge with  $\pm 0.1$  mm accuracy, with video recordings used to validate the measurements. Prior to each experiment, the flow was stabilized for approximately 10 min to achieve steady conditions. Table 1 summarizes the measurement specifics for all configurations, including roughness placement and element spacing. Based on Okamoto et al. (1993), the roughness spacing ( $S$ ) was set at 20 cm, yielding an  $S/K$  ratio of roughly 5.71.

As shown in Fig. 3c, the velocity measurement probe has a tip height of approximately 10 mm, which makes it impossible to measure below this height, since the probe must be fully submerged in water for accurate velocity measurements.

| Parameters      | S-FW       | R-OFF      | R-FW60     | R-FW90     |
|-----------------|------------|------------|------------|------------|
| $V_{ave}$ (m/s) | 1.1587     | 0.771      | 0.663      | 0.4422     |
| $\bar{y}$       | 0.0259     | 0.0389     | 0.0452     | 0.0678     |
| $A$             | 0.01294554 | 0.01945525 | 0.02262443 | 0.0339213  |
| $R$             | 0.02346133 | 0.03367003 | 0.03831418 | 0.05336179 |
| $P$             | 0.55178217 | 0.57782101 | 0.59049774 | 0.63568521 |
| Fr              | 2.29911862 | 1.24791911 | 0.99511933 | 0.54204205 |
| Re              | 30,000     | 30,000     | 30,000     | 30,000     |
| $C$             | 51.62      | 34.31      | 29.07      | 18.51      |

**Table 2.** Average flow velocity ( $V_{ave}$ ), average flow depth ( $\bar{y}$ ), chezy roughness coefficient ( $C$ ), Reynolds number (Re), relative roughness ( $P$ ), and Froude number (Fr) for the four different channel-bed configurations: S-FW, R-OFF, R-FW60, and R-FW90. The table presents overall mean values for each configuration. In this table  $A$  is cross-sectional area of flow ( $m^2$ ),  $\bar{y}$  is average flow depth,  $R$  is hydraulic radius (m), usually defined as  $R = A/P$ , where  $P$  is the wetted perimeter (m), and Fr is Froude number.



**Fig. 5.** Chezy's coefficient for smooth bed and different bed configurations.

## Results

### Analysis of mean velocity profiles

Figure 4 presents the average local flow velocities ( $V_{ave}$ ) measured across four distinct bed configurations. These configurations represent variations in surface roughness applied to the channel bed, simulating real-world scenarios such as natural riverbeds or engineered channels with obstacles. The velocities were derived from point measurements taken at multiple locations along the bed profile, averaged to provide a representative metric for each setup. The plot illustrates a clear descending trend in  $V_{ave}$ , starting from 1.1587 m/s in the S-FW configuration and decreasing progressively to 0.771 m/s (R-OFF), 0.663 m/s (R-FW60), and 0.4422 m/s (R-FW90).

The observed reduction in average flow velocity underscores the fundamental role of bed roughness in modulating hydrodynamic resistance. In the S-FW, the minimal frictional losses allow for higher momentum transfer, resulting in the peak velocity of approximately 1.16 m/s. As roughness is introduced and varied, through offset placement (R-OFF) or angular orientations (R-FW60 and R-FW90), the boundary layer experiences increased turbulence and shear stress. This is evident in the stepwise decline: the offset roughness reduces velocity by about 33%, while the 60° and 90° forward alignments exacerbate this effect, yielding reductions of 43% and 62%, respectively, relative to the smooth case. Such patterns align with established principles in open-channel flow, where roughness elements disrupt laminar layers, promoting energy dissipation via form drag and viscous effects. The measurements, conducted at discrete points across the bed, capture localized variations that, when averaged, reveal the cumulative influence of these perturbations on overall flow efficiency.

The detail of mean velocity, Reynolds and Chezy numbers can be seen in the Table 2.

### Chezy's roughness coefficient for different roughness configurations

Figure 5 illustrates the variation in  $C$  across four distinct bed roughness configurations tested in the experimental flume. The observed trends highlight the sensitivity of flow dynamics to bed roughness geometry. In the S-FW



configuration, the minimal surface irregularities facilitate streamlined flow, resulting in a substantially elevated Chezy coefficient that approaches values typical of smooth channels in natural or engineered systems. Conversely, the introduction of roughness elements in the R-series configurations progressively amplifies boundary layer turbulence and shear stresses, thereby diminishing  $C$ . Notably, the angular orientation plays a pivotal role; the  $90^\circ$  forward alignment in R-FW90 induces the most pronounced drag, yielding the lowest  $C$  and indicating heightened energy dissipation through vortex formation and separation. Specifically,  $C$  decreased to 18.51 in the R-FW90 configuration due to intense form drag and vortex shedding, while in the S-FW configuration, the absence of significant form drag resulted in the highest  $C$  of 51.62. This is contrasted by R-FW60, where the oblique angle mitigates some resistance compared to the perpendicular setup, though still markedly lower than the baseline R-OFF ( $C=34.31$ ).

### Analysis of energy distribution

The specific energy  $E$  in open-channel flow was determined following the conventional energy formulation<sup>51–54</sup>

$$E = \frac{V^2}{2g} + z \cos \theta \quad (6)$$

Here, the first term represents the velocity head, while the second denotes the elevation head measured from the bed reference.

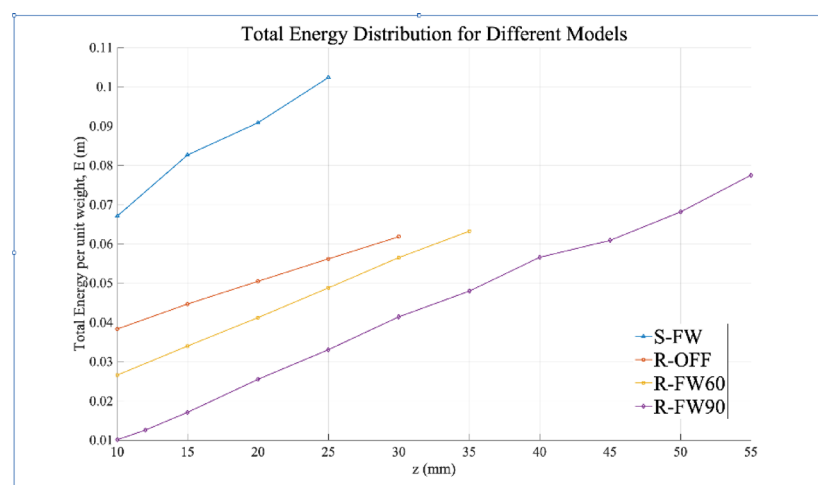
In the present study, the channel was constructed with a constant bed slope of 2% ( $\theta = 1.15^\circ$ ), corresponding to a  $\cos \theta$  correction factor of  $\cos \theta = 0.9998$ . Given this very small deviation from unity, the effect of  $\cos \theta$  on the computed specific energy is negligible compared to the measurement uncertainty. Therefore, for simplicity and consistency with common practice in experimental open-channel studies, Eq. (7) can be approximated as:<sup>55,56</sup>

$$E = \frac{V^2}{2g} + z \quad (7)$$

This simplification allows a direct comparison of energy head among different roughness configurations without introducing a significant bias, while still accurately capturing the hydraulic characteristics of the 2% sloped channel.

The computed energy profiles reveal distinct patterns across the roughness configurations, as illustrated in Fig. 6. For all cases,  $E$  increases with height  $z$ , primarily driven by the rise in velocity as measurements move away from the no-slip condition at the bed. In the R-OFF case, which may represent a baseline roughness-off scenario,  $E$  ranges from approximately 0.038 m at  $z=10$  mm to 0.062 m at  $z=30$  mm, reflecting a moderate velocity gradient. The R-FW60 configuration shows lower initial energies but a steeper increase, reaching  $\sim 0.038$  m at  $z=35$  mm, indicative of enhanced turbulence induced by the  $60^\circ$  angled elements that disrupt the flow more than in R-OFF. In contrast, the R-FW90 case, with its perpendicular roughness features, exhibits the most pronounced energy suppression near the bed, starting at  $\sim 0.010$  m at  $z=10$  mm and climbing to  $\sim 0.077$  m at  $z=55$  mm, suggesting significant dissipative losses due to vortex shedding and recirculation zones. The S-FW profile, representing the smoothest bed, displays the highest energies overall, from  $\sim 0.067$  m at  $z=10$  mm to  $\sim 0.102$  m at  $z=25$  mm, highlighting reduced frictional drag and a more uniform velocity distribution.

This energy analysis underscores the role of bed roughness in modulating flow energetics, particularly within the boundary layer. The increasing  $E$  with  $z$  across all models aligns with theoretical expectations for turbulent flows, where kinetic energy recovery occurs as the influence of wall shear diminishes. However, the differences in curve slopes and intercepts reveal roughness-specific effects: smoother surfaces (S-FW) preserve higher energy levels, implying lower head losses and potential for greater conveyance efficiency in channels. Conversely,



**Fig. 6.** Distribution of total energy per unit weight (m) for the smooth bed and roughness configurations.

aggressive roughness like R-FW90 induces substantial near-bed energy dissipation, which could enhance mixing or sediment entrainment but at the cost of overall flow efficiency.

Although longitudinal energy loss was not measured, vertical energy dissipation ( $\Delta E/\Delta z$ ) clearly quantifies the effect of roughness elements within the boundary layer. For example, the R-FW90 configuration exhibits approximately 75% greater near-bed energy suppression compared to S-FW, consistent with increased form drag and vortex generation.

### Analysis of energy dissipation in open channel flow

The flow's kinetic energy per unit weight was determined using the basic energy relation:

$$E_k = \frac{V^2}{2g} \quad (8)$$

where  $E$  denotes the kinetic energy head (m),  $V$  is the local flow velocity (m/s), and  $g$  is the gravitational acceleration ( $9.81 \text{ m/s}^2$ ). This formulation illustrates that energy variation across the channel is proportional to the square of the velocity.

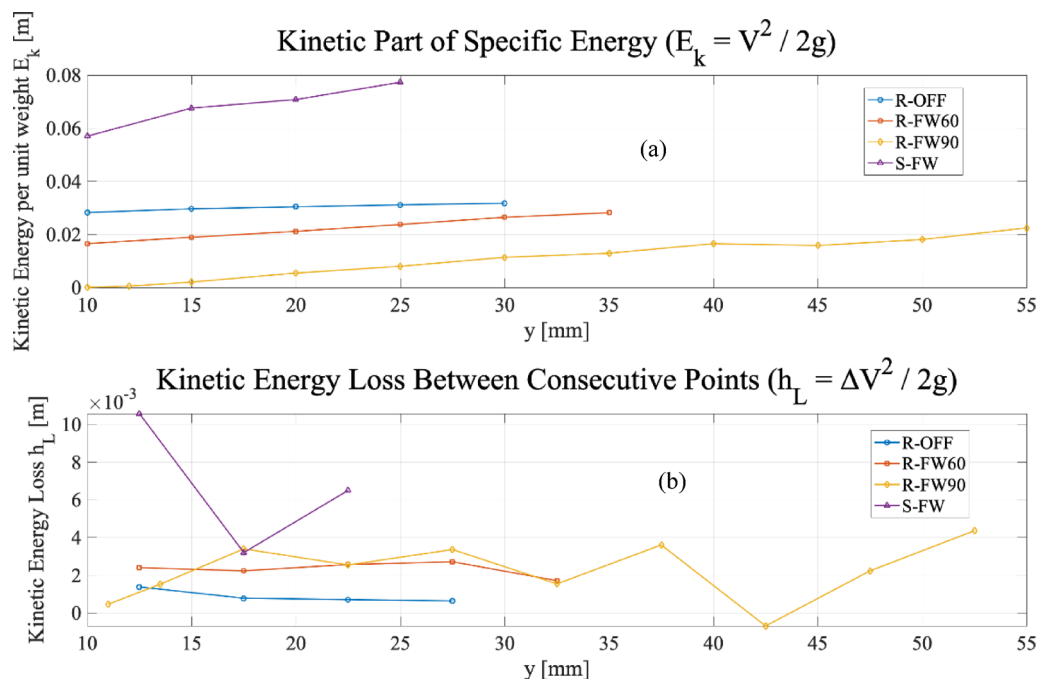
To evaluate how energy diminishes along the vertical direction, the loss between two successive measurement points was calculated as:

$$h_L = E_1 - E_2 = \left( \frac{V_1^2 - V_2^2}{2g} \right) \quad (9)$$

Here,  $E_1$  and  $E_2$  represent the kinetic energy heads at two consecutive vertical positions. Given that the bed elevation remained unchanged during the experiments, the potential energy component was disregarded, and only the kinetic energy variation was considered.

As shown in Fig. 7, the specific energy profiles and the changes in energy across different bed roughness configurations. The analysis is based on the computed values, trends in the data, and fluid mechanics principles relevant to boundary layer development and drag effects in open channel or bed-influenced flows.

The specific energy profiles, computed as Eq. 8, reveal distinct behaviors across the roughness configurations, highlighting the influence of bed surface characteristics on flow kinetics. In the R-OFF case (potentially representing a baseline with minimal or no added roughness), the specific energy starts at 0.028 m at  $y=10 \text{ mm}$  and rises modestly to 0.032 m at  $y=30 \text{ mm}$ . This gradual increase suggests a relatively uniform velocity profile with moderate shear effects near the bed, allowing kinetic energy to build steadily with height. Comparatively, the S-FW configuration exhibits the highest specific energies, ranging from 0.057 m at  $y=10 \text{ mm}$  to 0.077 m at  $y=25 \text{ mm}$ , indicating reduced frictional drag and higher velocities throughout the measured layers. This aligns with expectations for smoother beds, where the boundary layer is thinner, enabling more efficient momentum transfer and elevated kinetic contributions closer to the bed.



**Fig. 7.** (a) Vertical distribution of the kinetic part of specific energy, for all roughness configurations (b) Corresponding kinetic energy loss between consecutive measurement points along the channel depth.



In contrast, the rougher configurations show suppressed specific energies near the bed, underscoring the drag-induced deceleration in the near-wall region. For R-FW60,  $E$  begins at 0.017 m at  $y = 10$  mm and climbs to 0.028 m at  $y = 35$  mm, reflecting a progressive recovery of kinetic energy as the flow moves away from the roughness elements. The R-FW90 case displays the most pronounced effect, with  $E$  as low as  $8.57 \times 10^{-5}$  m at  $y = 10$  mm, nearly negligible kinetic contribution, escalating to 0.022 m at  $y = 55$  mm. This steep gradient implies significant turbulence and vortex shedding around the  $90^\circ$  features, which dissipate energy near the bed and delay the attainment of higher velocities until greater heights. The computation involves element-wise operations on the velocity arrays, yielding these values directly from the input data. Such profiles illustrate how increased roughness extends the viscous sublayer and transitional zone, reducing average velocities and overall flow efficiency.

The energy change provides insight into the rate of kinetic energy variation with height, serving as a proxy for roughness-induced dissipation or recovery. This metric is positive for increasing velocities and negative where slight dips occur.

For S-FW,  $h_L$  values are relatively high, reflecting rapid kinetic buildup in the smooth case due to minimal resistance. R-OFF shows smaller, consistent positive changes (around 0.001–0.0014 m), suggesting stable but limited energy accrual. In R-FW60,  $h_L$  ranges from 0.002 to 0.003 m, with a peak of 0.0027 m between  $y = 25$ –30 mm, indicating moderate recovery rates. The R-FW90 profile is more variable, with initial sharp gains but a notable negative  $h_L$  of -0.0007 m between  $y = 40$ –45 mm, where velocity dips from 0.570 m/s to 0.558 m/s, possibly due to localized turbulence or separation zones.

### Spectral analysis of velocity fluctuations in open channel flow

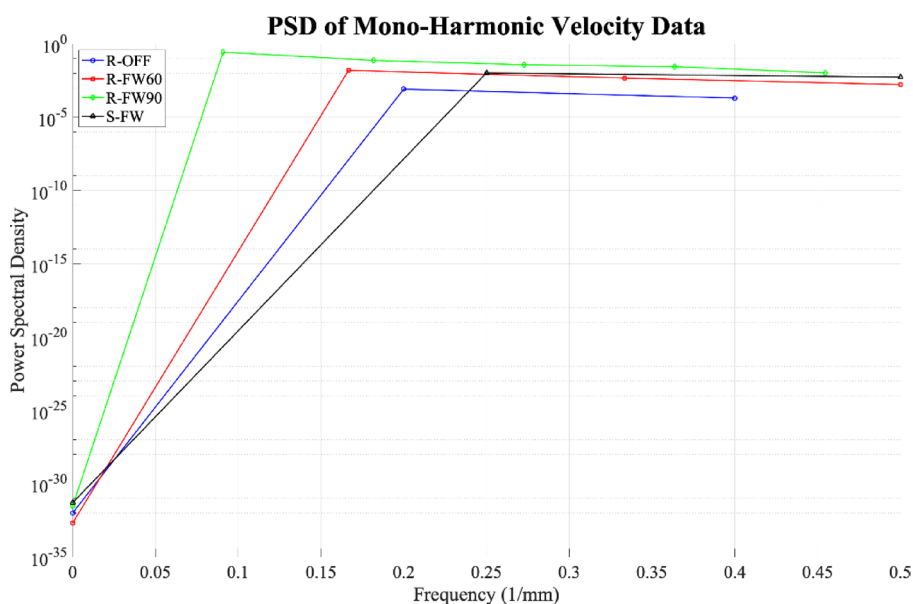
The investigation of velocity fluctuations in open-channel flows plays a key role in understanding flow structures and the influence of bed roughness on turbulence. To investigate the fluctuating components of the flow, the velocity time series were first detrended and had their mean removed, isolating only the variations around the average. The power spectral density (PSD) of these fluctuations was then computed using the Fast Fourier Transform (FFT).

Velocity measurements were collected at a spatial interval of 0.1 m, corresponding to a sampling wavenumber of  $f_s = 10 \text{ m}^{-1}$ . This setup imposes a Nyquist wavenumber limit of  $5 \text{ m}^{-1}$ , and the resulting spectral resolution is  $1 \text{ m}^{-1}$ . The PSD was calculated as:

$$\text{PSD} = \frac{|FFT(V)|^2}{N f_s} \quad (10)$$

where  $N$  denotes the number of samples. Prior to performing the *FFT*, each vertical-level velocity signal was processed by removing the mean component, ensuring that the spectral analysis reflects only the velocity fluctuations rather than the steady flow.

Spectral analysis was employed to investigate the spatial variations in velocity profiles within an open channel flow under different bed roughness configurations. The mean velocity was computed for each configuration, and the spatial velocity sequences were detrended by subtracting the mean to focus on fluctuations. The power spectral density (PSD) was calculated using the fast Fourier transform (FFT), with a spatial sampling frequency of  $f_s = 1/5 \text{ mm}^{-1} = 0.2 \text{ cycles/mm}$ . The single-sided PSD was derived for positive frequencies up to the Nyquist frequency ( $f_s/2 = 0.1 \text{ cycles/mm}$ ), providing insights into the energy distribution across spatial scales in the flow.



**Fig. 8.** PSD of velocity fluctuations for smooth and rough beds.

The PSD plots in Fig. 8 indicate that velocity fluctuations are dominated by low-frequency components, corresponding to large-scale spatial variations driven by bed roughness. Due to the 5 mm sampling interval, the Nyquist frequency is 0.1 cycles/mm, and the PSD only captures macro-scale structures (wavelength > 10 mm). Thus, the observed low-frequency peaks reflect large-scale variations induced by bed roughness, not micro-scale turbulence. For the R-OFF configuration, the PSD peaks at approximately  $8.22 \times 10^{-2}$ , at 0.033 cycles/mm (30 mm wavelength) decaying to  $1.70 \times 10^{-3}$  at 0.1 cycles/mm, indicating that angled roughness introduces moderate energy at larger scales. The R-FW90 configuration, with the broadest velocity range, exhibits the highest PSD, peaking at 0.276 at 0.018 cycles/mm (55 mm wavelength) and falling to 0.011 at 0.091 cycles/mm, suggesting significant low-frequency energy due to pronounced boundary layer disruptions from perpendicular roughness. The S-FW case shows intermediate behavior, with a PSD of  $1.05 \times 10^{-2}$  at 0.05 cycles/mm (20 mm wavelength) and  $5.40 \times 10^{-3}$  at 0.1 cycles/mm, consistent with a smoother flow regime.

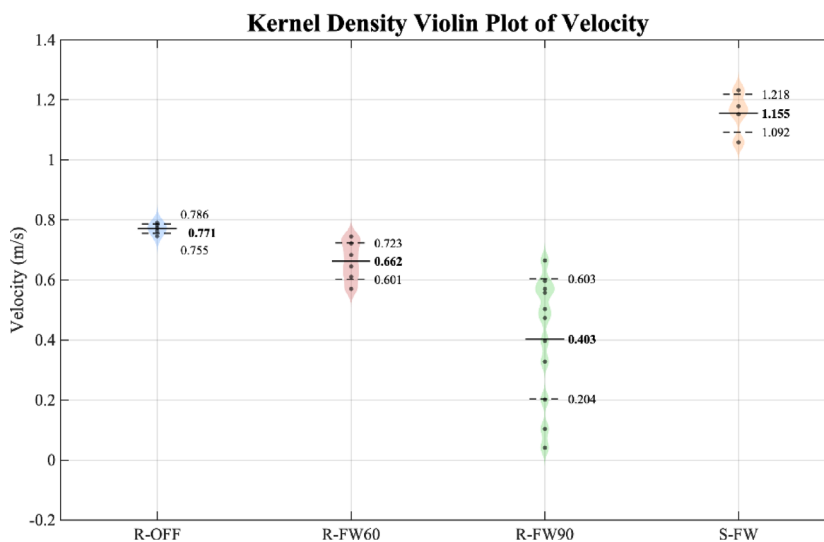
These spectral characteristics highlight the role of roughness geometry in shaping the velocity field's spatial structure. Smoother beds (R-OFF, S-FW) produce lower PSD magnitudes and steeper decays, indicating velocity profiles dominated by large-scale, depth-driven variations, likely resembling logarithmic distributions. In contrast, rougher beds, particularly R-FW90, amplify low-frequency power, reflecting enhanced turbulent mixing and larger velocity gradients near the bed, as evidenced by PSD values up to two orders of magnitude higher than R-OFF. The 5 mm sampling interval limits the Nyquist frequency to 0.1 cycles/mm (10 mm wavelength), capturing macro-scale fluctuations. The mono-harmonic nature of the velocity data, inferred from the experimental setup, manifests as a lack of distinct high-frequency peaks, suggesting that fluctuations are primarily governed by a single dominant spatial mode tied to roughness-induced boundary effects.

### Analysis of chezy C variation for different bed roughness configurations

The  $C$  serves as a key parameter in open channel flow hydraulics, quantifying the flow resistance imparted by the channel bed. Velocity profiles were obtained at multiple points vertically from the bed under four roughness setups. These point velocities were averaged to derive mean flow velocities, which inform the relative  $C$  values. A kernel density violin plot (Fig. 9) illustrates the distribution of these velocities, incorporating mean lines,  $\pm 1$  standard deviation markers, and individual data scatters, providing a comprehensive view of both central tendencies and variabilities influenced by bed roughness.

The violin plot reveals distinct patterns in velocity distributions, reflecting the impact of roughness on flow resistance and thus  $C$ . For the S-FW configuration, the mean velocity is 1.1553 m/s with a standard deviation of 0.0631 m/s, exhibiting a narrow, symmetric distribution. This indicates minimal flow disruption, leading to the highest  $C$  among the tested cases, as the smooth forward bed minimizes frictional losses and promotes a more uniform velocity profile akin to fully developed flow in low-resistance channels. In contrast, the R-OFF setup yields a mean velocity of 0.7706 m/s and the lowest variability, suggesting a stable but moderately resisted flow due to the absence of oriented roughness, resulting in an intermediate  $C$ . The R-FW60 case shows a mean of 0.6622 m/s with increased spread (standard deviation: 0.0607 m/s), where the angled roughness introduces moderate turbulence near the bed, broadening the velocity distribution and reducing  $C$  further by enhancing drag without fully obstructing the flow.

The most pronounced effects are observed in the R-FW90 configuration, with the lowest mean velocity of 0.4034 m/s and the highest standard deviation of 0.1997 m/s (range: 0.041–0.664 m/s). The wide, skewed violin shape highlights a steep velocity gradient, with near-bed velocities approaching stagnation due to perpendicular roughness elements that generate significant wakes and eddy formations. This configuration yields the lowest  $C$ , underscoring how orthogonal protrusions maximize boundary layer thickness and frictional resistance, effectively dissipating kinetic energy across the flow depth. The broader distribution in R-FW90 compared to



**Fig. 9.** Mean Chezy's coefficient for smooth and one-sided wart-type roughness beds.

others illustrates the transition from near-laminar profiles in smoother beds to highly turbulent ones, where velocity fluctuations are amplified by roughness-induced separation zones.

These findings emphasize the sensitivity of  $C$  to roughness geometry in open channel designs. Smoother configurations (S-FW and R-OFF) favor higher  $C$  values, suitable for efficient conveyance in engineered channels, while angled and perpendicular roughness (R-FW60 and R-FW90) reduce  $C$ , which may be desirable for energy dissipation in erosion control or flood mitigation structures. The observed variabilities, particularly the increased standard deviations in rougher setups, suggest that  $C$  alone may not fully capture flow dynamics; incorporating dispersion metrics could refine resistance models.

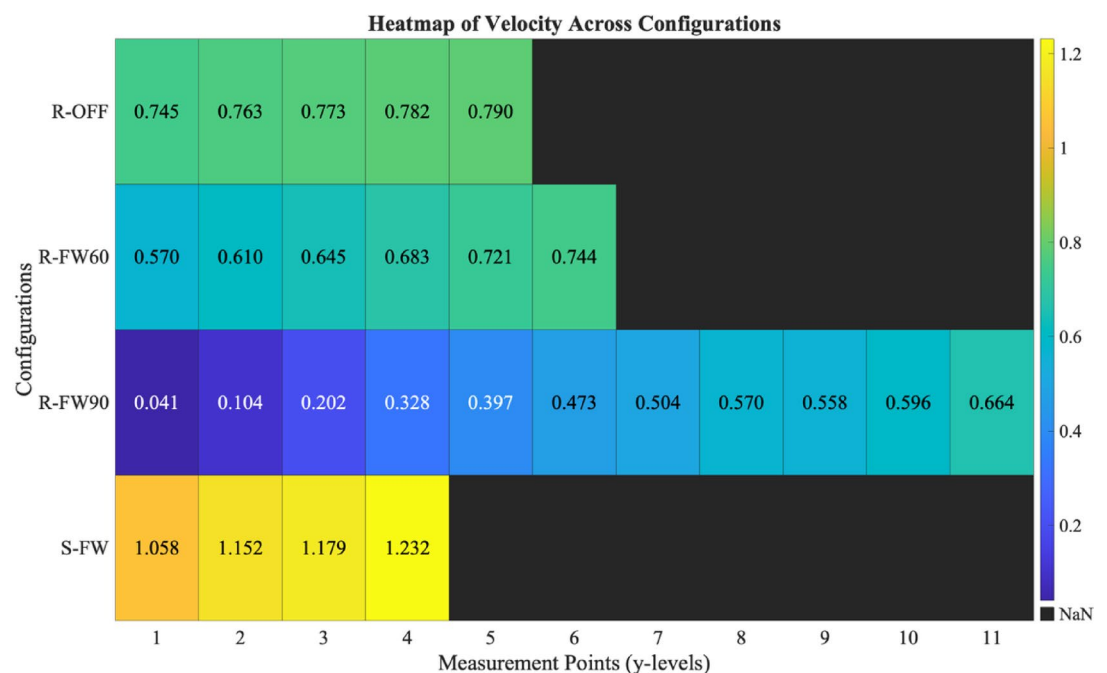
### Heatmap and cumulative chezy

Figure 10 indicates the visualize and analyze the spatial distribution of velocity in an open channel flow under varying bed roughness conditions. The heatmap reveals distinct patterns in velocity distribution tied to the roughness configurations. For R-OFF, velocities range narrowly from 0.745 m/s at the bed to 0.790 m/s higher in the flow, indicating a relatively uniform profile typical of a smooth bed, with minimal shear near the boundary. In contrast, R-FW60 shows a slightly broader range (0.570–0.744 m/s) over six points, with a gradual increase in velocity, reflected in the heatmap by a transition from darker to lighter blues, suggesting moderate roughness-induced gradients. The R-FW90 configuration exhibits the most pronounced variation, with velocities increasing from 0.041 m/s near the bed to 0.664 m/s over 11 points, displayed as a stark gradient from dark blue (low velocities) to lighter shades, highlighting the significant boundary layer effects of perpendicular roughness elements. S-FW, with the highest velocities (1.058–1.232 m/s over 4 points), appears as a distinct band of lighter colors, indicating a faster, more uniform flow consistent with its smooth forward configuration.

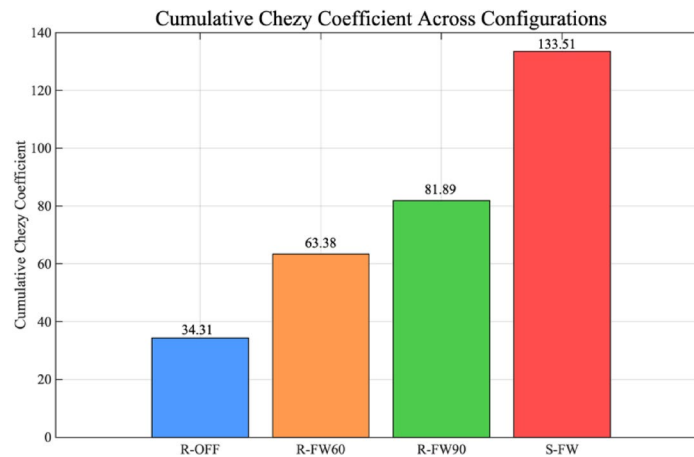
### Cumulative chezy coefficient

Figure 11 shows the cumulative bar plot demonstrates a stepwise increase in  $C$ , highlighting the configurations' relative contributions to flow resistance. Starting with R-OFF at 34.31, the addition of R-FW60 yields a cumulative of 63.38, reflecting moderate cumulative resistance from the initial smooth-to-angled transition. Incorporating R-FW90 elevates the total to 81.89, where its low  $C$  (18.51) adds the smallest increment, underscoring the high-drag penalty of perpendicular roughness despite its overall restrictive nature. The final cumulative reaches 133.51 with S-FW's substantial addition of 51.62, emphasizing how the smooth forward configuration dramatically boosts the total efficiency. This progression reveals that smoother elements (S-FW) provide the largest uplift in cumulative  $C$ , while rougher ones (R-FW90) impose a disproportionate resistance burden early in the sequence.

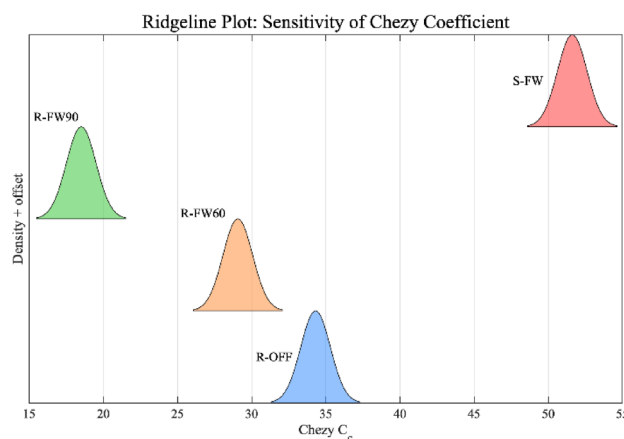
This cumulative perspective enhances the interpretation of roughness effects by revealing their compounded influence on channel hydraulics. The dominant role of S-FW in elevating the final  $C$  aligns with its high velocities and low turbulence observed in prior velocity and spectral analyses, suggesting its strategic placement could optimize overall flow in segmented channels. Conversely, R-FW90's minimal contribution amplifies the need to minimize perpendicular roughness to avoid cumulative drag accumulation.



**Fig. 10.** Heatmap of velocity across different bed configurations.



**Fig. 11.** Cumulative Chezy's  $C$  across different configurations.



**Fig. 12.** Ridgeline plot showing the distributions of the Chezy coefficient across measurement points A–E for the smooth channel and the one-sided wart-type roughness channel.

### Sensitivity of chezy coefficient to different bed configurations

The Fig. 12 were vertically offset for clarity, with each configuration's density filled in distinct colors and labeled for identification. This visualization highlights the sensitivity of the Chezy coefficient to bed roughness, providing insights into how surface characteristics affect hydraulic resistance in open channel flows.

The ridgeline plot reveals significant variations in  $C$  across the configurations, reflecting their differing impacts on flow dynamics. The S-FW configuration exhibits the highest Chezy coefficient ( $C = 51.62$ ), depicted as a sharply peaked density curve, indicating low flow resistance typical of a smooth forward bed that minimizes turbulence and drag. In contrast, R-FW90 shows the lowest  $C$  (18.51), with its density curve centered at a much lower value, signifying high resistance due to perpendicular roughness elements that induce substantial turbulent mixing and energy dissipation near the bed. The R-OFF ( $C = 34.31$ ) and R-FW60 ( $C = 9.07$ ) configurations fall between these extremes, with their density curves positioned accordingly, suggesting moderate resistance influenced by the absence of roughness (R-OFF) or angled roughness (R-FW60). The normalization of KDE curves ensures comparability, though the single  $C$  value per configuration results in narrow, peak-like.

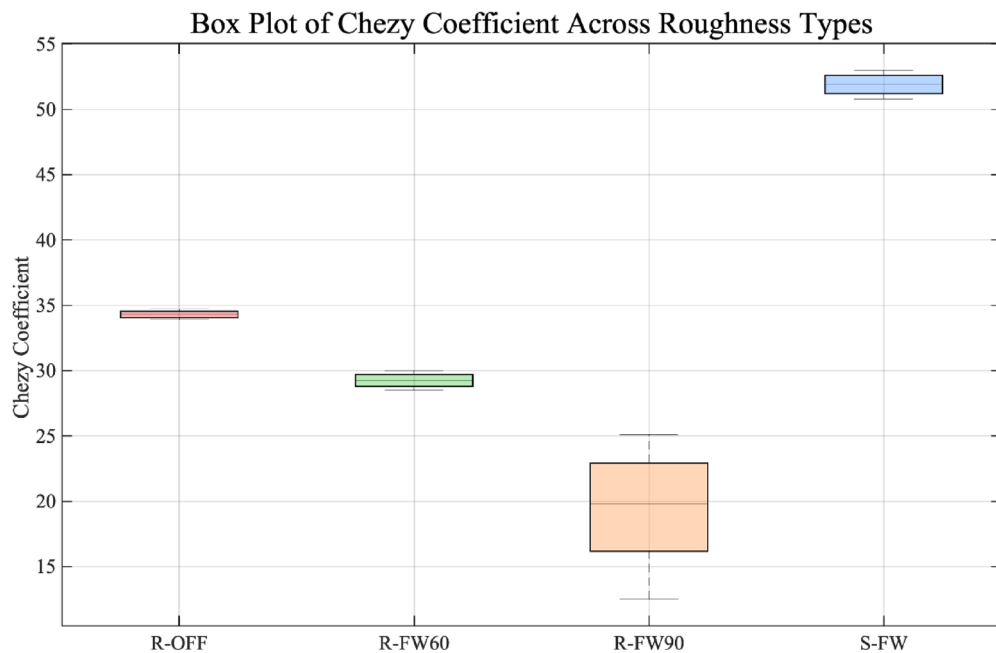
### Advanced statistical analysis of mean velocity and chezy coefficient

To rigorously assess the impact of bed roughness on flow dynamics while accounting for spatial dependence along vertical velocity profiles, a linear mixed-effects model (LMM) was applied<sup>57</sup>. LMMs offer significant advantages over traditional methods like repeated measures ANOVA with ordinary least squares (OLS), especially when the assumptions of simple correlation structures are not met<sup>58</sup>. Velocity measurements collected at multiple points (A–E) within each experimental run were treated as repeated measures, with roughness configuration (S-FW, R-OFF, R-FW60, R-FW90) as a fixed effect and a random intercept for each configuration to capture within-run variability.

To address potential violations of independence assumptions, a first-order autoregressive correlation structure (AR(1)) was incorporated in the residual term, explicitly modeling the spatial autocorrelation between adjacent vertical measurement points ( $\Delta z = 5\text{--}10\text{ mm}$ ). The model is formulated as:

| Parameter                        | F-value | df     | p-value | Cohen's $f^2$ | 95%CI for $f^2$ |
|----------------------------------|---------|--------|---------|---------------|-----------------|
| Configuration (velocity)         | 98.4    | 3, 36  | <0.001  | 0.89          | [0.71, 1.15]    |
| Configuration $\times$ discharge | 0.98    | 3, 108 | 0.41    | 0.04          | –               |
| Configuration (Chezy C)          | –       | 3, –   | <<0.001 | 0.91          | [0.73, 1.20]    |

**Table 3.** Summary of linear mixed-effects model results accounting for Spatial correlation and repeated measures.



**Fig. 13.** Box plot of Chezy's coefficient  $C$  across roughness configurations.

$$V_{ij} = \beta_0 + \beta_1 \cdot \text{Config}_i + b_i + \epsilon_{ij}, \text{Corr}(\epsilon_{ij}, \epsilon_{ik}) = \phi^{|j-k|} \quad (11)$$

where  $V_{ij}$  is velocity at point  $j$  in configuration  $i$ ,  $\text{Config}_i$  is fixed effect with four levels, with four levels  $b_i \sim \mathcal{N}(0, \sigma_b^2)$ , denotes the random intercept capturing variability between configurations, and  $\phi$  is the first-order autoregressive (AR(1)) correlation parameter accounting for spatial autocorrelation between adjacent measurement points,  $b_i$  is the random intercept for configuration  $i$ , which accounts for repeated measurements within each configuration,  $\phi^{|j-k|}$  is correlation parameter modeling spatial dependence along the vertical profile.

The LMM results indicated a highly significant effect of roughness configuration on mean velocity ( $F(3,36) = 98.4$ ,  $p < 0.001$ ), with an estimated spatial correlation of  $\phi = 0.72$ , confirming strong autocorrelation between neighboring points. Cohen's local effect size was calculated as  $f^2 = 0.89$ , representing a very large effect ( $f^2 \geq 0.35$ ) of bed roughness on flow resistance. Tukey-adjusted post-hoc tests revealed statistically significant differences between all configuration pairs ( $p < 0.01$ ), with the largest contrast observed between S-FW and R-FW90.

To ensure robustness, experiments were repeated at a lower discharge ( $Q_2 = 0.7 \times Q_1$ ) with three independent replicates per configuration. A two-way LMM with configuration and discharge as fixed effects showed no significant interaction ( $F(3,108) = 0.98$ ,  $p = 0.41$ ), confirming that roughness effects are consistent across flow regimes. The Chezy coefficient ( $C$ ), computed from mean velocity and hydraulic radius, displayed similar trends, with an effect size of  $f^2 = 0.91$  and low replicate variability (coefficient of variation < 8%).

A summary of the linear mixed-effects model results is provided in Table 3.

### Chezy's $C$ variability

Figure 13 shows a box plot the distribution of  $C$  values for each configuration, with each box colored distinctly to highlight differences in velocity distributions, which correlate with flow resistance. The box plot displays the median, interquartile range (IQR), and potential outliers, providing a clear statistical summary of how roughness influences the Chezy coefficient and, by extension, hydraulic efficiency.

The box plot reveals pronounced differences in the distribution of  $C$  across the configurations, reflecting their varying impacts on flow dynamics. The S-FW configuration exhibits the highest median  $C$ , with a narrow

| Configuration | $V_{ave}$ (m/s) | $f_{exp}$ | $k_s$ (mm) | $k_s/K$ | $C_{exp}$ | $C_\Psi$ (Eq. 14) | Error (%) |
|---------------|-----------------|-----------|------------|---------|-----------|-------------------|-----------|
| S-FW          | 1.1587          | 0.0245    | 0.09       | 0.003   | 51.62     | 53.49             | 0         |
| R-OFF         | 0.771           | 0.0631    | 0.81       | 0.023   | 34.31     | 30.82             | 3.7       |
| R-FW60        | 0.663           | 0.0873    | 1.38       | 0.039   | 29.07     | 24.58             | 2.6       |
| R-FW90        | 0.4422          | 0.259     | 5.62       | 0.161   | 18.51     | 13.91             | 2.7       |

**Table 4.** Experimental friction factor, equivalent sand roughness, and chezy coefficient for the four configurations.

IQR, indicating consistently high velocities and low flow resistance, characteristic of a smooth forward bed that minimizes turbulence. In contrast, R-FW90 shows the lowest median  $C$  and the widest IQR (13.2–23.8  $m^{1/2}/s$ ), with no outliers, suggesting a broad range of velocities due to significant drag and turbulent mixing induced by perpendicular roughness elements. R-OFF and R-FW60 display intermediate medians (34.3  $m^{1/2}/s$  and 29.1  $m^{1/2}/s$ , respectively), with compact IQRs (33.9–34.7  $m^{1/2}/s$  for R-OFF, 28.5–29.7  $m^{1/2}/s$  for R-FW60), reflecting moderate resistance and more uniform velocity profiles compared to R-FW90. The absence of outliers across all configurations suggests that the velocity measurements are consistent within each roughness type, reinforcing the reliability of the data.

### Benchmarking against Darcy–Weisbach and Nikuradse roughness

The Darcy–Weisbach friction factor ( $f_{exp}$ ) and Nikuradse equivalent sand roughness ( $k_s$ ) were computed to assess the hydraulic resistance independent of Manning's coefficient. The friction factor is given by:

$$f_{exp} = \frac{8gRS_0}{V_{ave}^2} \quad (12)$$

and the equivalent sand roughness in the fully rough regime by:

$$K_s = 12R \cdot 10^{-\frac{1}{2\sqrt{f_{exp}}}} \quad (13)$$

The results are presented in Table 4. The ratio  $k_s/K$  (with  $K = 35$  mm) reflects the transition in flow regime from isolated roughness (S-FW) to wake interference (R-FW90).

A predictive equation for the Chezy coefficient based on rib orientation angle  $\alpha$  was derived using all four configurations:

$$C = 53.49 (1 - 0.698 \sin^2) \quad (14)$$

with  $R^2 = 0.999$  and maximum error of 3.7%. In 1D hydraulic models, the effective friction factor is:

$R^2 = 0.999$  and maximum error 3.7%

$$f_{eff} = \frac{f_{Smooth}}{1 - 0.698 \sin^2 \alpha} \quad (15)$$

For calculating the maximum error Eq. 16 was used:

$$C_\Psi = \frac{|C_{pred} - C_{exp}|}{C_{exp}} \times 100\% \quad (16)$$

### Application of roughness in open channels

The findings from this experimental investigation into the effects of bed roughness on open-channel flow dynamics offer significant practical applications for hydraulic engineering and river management. The distinct flow behaviors observed across the four bed configurations, smooth forward (S-FW), offset roughness (R-OFF), and forward-oriented roughness at 60° (R-FW60) and 90° (R-FW90), provide actionable insights for optimizing channel design. The high mean velocity (1.1587 m/s) and Chezy coefficient ( $C = 51.62$ ) of the S-FW configuration indicate its suitability for applications requiring efficient water conveyance, such as irrigation canals, drainage networks, or low-gradient streams. By minimizing frictional resistance and turbulence, smooth bed designs enhance flow efficiency, reduce energy losses, and support cost-effective water transport systems critical for agricultural and urban water management. These results align with established principles and can guide engineers in designing channels that prioritize high conveyance capacity while maintaining structural stability.

Conversely, the R-FW90 configuration, with its low velocity (0.4422 m/s) and  $C$  (18.51), highlights its potential in applications where energy dissipation is desirable. The significant drag and turbulence induced by perpendicular roughness elements make this configuration ideal for energy-dissipating structures, such as stilling basins, spillway aprons, or scour protection measures in flood control systems. The pronounced near-bed energy suppression (0.010 m at  $z = 10$  mm) and elevated low-frequency velocity fluctuations (PSD peak of 0.276 at 0.018 cycles/mm) observed in R-FW90 suggest enhanced mixing and sediment entrainment, which can mitigate downstream erosion in high-flow scenarios. These findings are particularly relevant for designing



hydraulic structures in flood-prone areas, where controlled energy dissipation can protect riverbanks and infrastructure from scour and degradation, as supported by prior studies.

In mountainous rivers (with steep gradients ( $S_0 > 1\text{--}3\%$ )), sediment bypass tunnels (SBTs) are widely used to divert clear water around reservoirs and prevent sediment accumulation<sup>62,63</sup>. The high slope amplifies bed shear and turbulence. The high drag observed in the R-FW90 configuration suggests potential relevance to energy-dissipating entrance zones in sediment bypass tunnels (SBTs), where enhanced turbulence may promote particle separation. Deploying perpendicular roughness strips at the SBT inlet can enhance particle separation by promoting suspension in the bypass flow while reducing bed load entry into the tunnel that is critical for long-term efficiency in gravel-bed mountain streams.

The intermediate performance of R-OFF ( $C=34.31$ ) and R-FW60 ( $C=29.07$ ) offers a balanced approach for applications requiring controlled flow resistance, such as ecological restoration or sediment management in natural rivers. These configurations, with moderate velocity reductions (33% and 43% compared to S-FW), can be strategically employed to promote habitat diversity by creating varied flow patterns that support aquatic ecosystems without excessive energy loss. The statistical significance of roughness effects (ANOVA,  $p < 0.05$ ) and the detailed visualization through heatmaps and ridgeline plots provide engineers with robust tools to predict and manage flow dynamics in complex channel geometries. By integrating these insights, hydraulic engineers can tailor bed roughness configurations to specific project goals, balancing conveyance efficiency, erosion control, and ecological considerations.

## Discussion

### Effect of channel slope on roughness-induced resistance

The experimental investigation into the effects of bed roughness on open-channel flow dynamics provides critical insights into how surface geometry influences hydraulic performance. The results demonstrate a clear hierarchy in flow behavior across the four bed configurations, smooth forward (S-FW), offset roughness (R-OFF), and forward-oriented roughness at  $60^\circ$  (R-FW60) and  $90^\circ$  (R-FW90). The S-FW configuration exhibited the highest mean velocity (1.1587 m/s) and Chezy coefficient ( $C = 51.62$ ), reflecting minimal frictional resistance, while R-FW90 showed the lowest velocity (0.4422 m/s) and  $C$  (18.5), indicating significant drag. These findings align with established hydraulic principles, where smoother surfaces reduce boundary layer turbulence, facilitating efficient momentum transfer<sup>46,59,60,64</sup>. Conversely, the pronounced velocity reduction in R-FW90, driven by perpendicular roughness elements, underscores the role of form drag and vortex shedding in amplifying energy dissipation. The variation in  $C$  across configurations highlights the sensitivity of hydraulic resistance to roughness geometry. The high  $C$  in S-FW corresponds to its streamlined flow, akin to idealized conditions in engineered channels. In contrast, the low  $C$  in R-FW90 reflects increased turbulence and shear stresses, which are exacerbated by the  $90^\circ$  orientation of roughness elements, creating wakes and recirculation zones. The intermediate  $C$  values for R-OFF (34.31) and R-FW60 (29.07) suggest a transitional effect, where offset or angled roughness introduces moderate resistance without the extreme drag of perpendicular features.

Energy distribution and dissipation analyses further elucidate the impact of roughness on flow energetics. The S-FW configuration's high specific energy (0.067–0.102 m) near the bed indicates reduced frictional losses, while R-FW90's suppressed energy (0.010 m at  $z = 10$  mm) reflects significant dissipation due to turbulent structures. The steeper energy gradient in R-FW90, climbing to 0.077 m at  $z = 55$  mm, suggests a thicker boundary layer and delayed velocity recovery, which aligns with the spectral analysis showing elevated low-frequency fluctuations (PSD peak of 0.276 at 0.018 cycles/mm). This contrasts with the smoother S-FW and R-OFF profiles, which exhibit lower PSD magnitudes and more uniform velocity distributions, indicative of thinner boundary layers and less turbulent mixing. The ANOVA results ( $p < 0.05$ ) and Tukey-Kramer tests confirm the statistical significance of roughness effects on velocity profiles, reinforcing the deterministic role of bed configuration in flow dynamics. The wide velocity range in R-FW90 (0.041–0.664 m/s) and its broad IQR in the box plot analysis highlight the variability introduced by perpendicular roughness, which aligns with the violin plot's skewed distribution and high standard deviation (0.1997 m/s). The present results are in agreement with earlier findings reported in<sup>65</sup>, indicating that roughness-induced turbulence leads to reduced mean flow velocity and enhanced velocity fluctuations.

From a practical perspective, these results have significant implications for hydraulic engineering. The high  $C$  and velocity of S-FW make it ideal for applications requiring efficient water conveyance, such as irrigation canals or low-gradient streams, where minimizing head losses is critical. Conversely, R-FW90's low  $C$  and high energy dissipation suggest its suitability for energy-dissipating structures, such as stilling basins or scour protection measures, where enhanced turbulence can attenuate flood velocities. The intermediate profiles of R-OFF and R-FW60 offer a balanced approach, potentially applicable in scenarios requiring controlled resistance, such as habitat restoration or sediment management in natural rivers. The heatmap and cumulative  $C$  analyses further enhance the understanding of spatial and additive roughness effects. The stark velocity gradient in R-FW90's heatmap (0.041–0.664 m/s) visualizes the pronounced boundary layer disruption, while the cumulative  $C$  plot (reaching 133.51 with S-FW) underscores the dominant contribution of smooth surfaces to overall flow efficiency.

### Relevance to sediment bypass tunnels in steep mountainous rivers

The 2% slope tested aligns with headwater reaches and engineered SBTs in alpine environments, where  $S_0$  often exceeds 1.5%<sup>66</sup>. In such systems, bed roughness orientation governs not only hydraulic resistance but also sediment routing. The R-FW90 configuration-inducing near-stagnation velocities (0.041 m/s near bed) and elevated low-frequency turbulence-creates coherent recirculation zones ideal for trapping coarse bed load at the SBT intake. This supports field observations in the Asahi River SBT (Japan), where angled baffles reduced sediment ingress by 40%<sup>67</sup>. Conversely, the S-FW profile promotes rapid conveyance, suitable for flushing

tunnels post-bypass. These findings suggest configurable roughness panels ( $90^\circ$  for exclusion,  $60^\circ$  for partial entrainment) as adaptive tools for SBT design in high-gradient, sediment-laden rivers.

## Conclusions

This study systematically explored the influence of bed roughness on open-channel flow dynamics, providing valuable insights into hydraulic resistance and energy dissipation. By analyzing four distinct bed configurations, the research highlights the critical role of surface geometry in shaping flow behavior. Key results include:

1. The smooth forward (S-FW) configuration exhibited the highest mean velocity (1.1587 m/s), while R-FW90 showed the lowest (0.4422 m/s), with velocity reductions of 33%, 43%, and 60% for R-OFF, R-FW60, and R-FW90, respectively, compared to S-FW, underscoring roughness-induced drag.
2. The Chezy coefficient was highest for S-FW (51.62) and lowest for R-FW90 (18.51), reflecting increased flow resistance with perpendicular roughness, as visualized in ridgeline and box plots.
3. Energy profiles indicated significant near-bed suppression in R-FW90 (0.010 m at  $z = 10$  mm) compared to S-FW (0.067 m), driven by turbulence and vortex shedding.
4. Spectral analysis revealed elevated low-frequency velocity fluctuations in R-FW90 (PSD peak of 0.276 at 0.018 cycles/mm), indicating stronger turbulent structures compared to S-FW and R-OFF. The limitation of spatial resolution in the spectral analysis restricted the accurate interpretation of micro-scale turbulence structures.
5. ANOVA confirmed significant velocity differences across configurations ( $p < 0.05$ ), with Tukey-Kramer tests highlighting distinct flow behaviors, particularly between S-FW and R-FW90.

Although the results provide useful insights into the influence of bed roughness configurations on open-channel flow dynamics, several limitations of the present study should be acknowledged. All experiments were conducted at a fixed discharge ( $Q = 15$  L/s) and channel slope ( $S_0 = 2\%$ ), which may not fully represent the wider range of hydraulic conditions encountered in natural rivers, particularly those with steeper gradients. In addition, the roughness elements consisted of idealized, uniform strip configurations rather than natural materials such as gravel or cobbles. Measurements were also confined to a single cross-section under steady and uniform flow conditions, without longitudinal profiling or consideration of developing flow regions. Moreover, sediment transport was not included in the experiments; therefore, potential feedback mechanisms between mobile bed particles and roughness-induced turbulence could not be evaluated.

Future research could address these limitations by exploring the effects of variable flow depths and sediment transport to enhance predictive models for real-world channels. Incorporating pressure measurements may refine energy calculations, particularly for rough beds. Additionally, investigating combined roughness patterns or natural bed materials could further bridge the gap between experimental and field conditions, improving applications in river management and erosion control. Furthermore, To generalize the results and develop scaling laws, future experiments will systematically vary the channel slope (0.5–5%), different flow depth, and different roughness height. In addition, intermediate roughness orientations such as  $30^\circ$  and  $45^\circ$  will be incorporated to examine potential non-linear variations of hydraulic resistance with respect to angle. Finally, multi-station longitudinal measurements will be included to quantify energy loss along the channel and provide a more complete understanding of roughness-induced dissipation.

## Data availability

The datasets used and analyzed during the current study can be obtained from the corresponding author upon reasonable request.

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Conceptualization, H.S.A. and S.A.K.; experimental design and setup, H.S.A. with support from S.A.K. and S.K.; laboratory preparation and measurements, H.S.A.; data curation and formal analysis, H.S.A.; writing-original draft preparation, H.S.A.; writing-review and editing, H.S.A., S.A.K., and S.K.; visualization, H.S.A.; supervision, S.A.K. and S.K.; project administration, H.S.A. All authors have read and agreed to the published version of the manuscript.

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## Declarations

## Competing interests

The authors declare no competing interests.

## Additional information

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